

SHENANDOAH, IA, USA

WMO: 725467

Lat: 40.751N

Lon: 95.413W

Elev: 296

StdP: 97.82

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04973

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-16.3	-23.0	0.5	-17.8	-21.1	0.6	-15.2	11.8	0.4	10.9	-0.5	1.9	0	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	35.2	26.1	33.7	25.2	32.4	24.7	28.0	33.2	26.8	32.0	25.7	30.7	5.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	23.5	32.1	25.2	21.0	30.4	24.0	19.6	29.0	91.2	33.6	86.1	32.2	80.8	30.6	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.6	8.5	DB	-25.5	36.6	3.9	1.8	-28.3	37.9	-30.6	38.9	-32.8	39.9	-35.7	41.2	(4)
(5)			WB	-26.1	29.4	3.7	1.3	-28.7	30.4	-30.9	31.2	-32.9	31.9	-35.6	32.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-3.4	-1.5	5.2	11.6	18.0	23.1	25.1	23.5	19.6	12.7	4.9	-1.3	(6)	
	DBStd	11.41	6.93	7.03	6.78	5.61	4.77	3.61	3.46	3.22	4.71	5.29	5.71	6.23	(7)	
	HDD10.0	1523	416	326	178	46	3	0	0	0	1	32	169	353	(8)	
	HDD18.3	3166	674	557	409	212	66	5	1	2	42	186	404	608	(9)	
	CDD10.0	2076	0	2	29	93	249	392	468	418	287	117	17	3	(10)	
	CDD18.3	678	0	0	2	9	54	147	212	162	78	12	1	0	(11)	
	CDH23.3	7147	0	0	21	138	557	1534	2343	1552	825	162	7	9	(12)	
	CDH26.7	2952	0	0	2	41	200	638	1075	624	319	47	1	6	(13)	
(14) Wind	WSAvg	3.6	3.6	3.8	4.5	4.7	4.2	3.5	2.6	2.3	2.9	3.5	3.7	3.5	(14)	
(15) Precipitation	PrecAvg	886	21	29	48	80	141	147	106	88	71	59	41	25	(15)	
	PrecMax	1337	50	81	112	209	294	373	363	235	178	212	111	80	(16)	
	PrecMin	662	2	3	0	20	40	56	0	25	9	0	0	0	(17)	
	PrecStd	169	13	19	33	40	71	71	78	54	42	50	32	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	19.6	26.0	32.0	34.9	36.1	37.8	36.1	35.1	31.2	24.0	18.6	(19)	
		MCWB	8.1	12.2	15.5	19.1	23.0	25.6	26.9	26.6	24.0	20.7	15.5	15.8	(20)	
	2%	DB	12.0	15.2	22.5	27.5	31.3	33.9	36.0	33.8	32.6	27.5	20.1	13.6	(21)	
		MCWB	7.1	9.8	14.4	17.1	21.4	24.4	27.4	25.9	23.2	17.8	13.6	9.5	(22)	
	5%	DB	8.1	12.1	19.0	24.2	28.7	32.4	33.8	32.3	30.2	24.9	17.2	10.9	(23)	
		MCWB	4.8	7.6	12.8	15.7	20.0	24.1	26.7	25.5	22.4	16.9	11.5	7.7	(24)	
	10%	DB	5.9	7.9	16.3	22.0	27.0	30.4	32.4	30.1	27.9	22.3	14.0	7.5	(25)	
		MCWB	3.2	4.9	11.3	14.6	19.1	22.9	26.0	24.6	20.8	15.6	9.2	4.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	13.8	17.9	21.0	25.3	27.9	29.9	28.7	25.7	22.9	16.9	15.6	(27)	
		MCDB	12.9	17.0	22.8	26.7	32.1	33.0	35.5	34.0	32.6	28.1	22.0	18.1	(28)	
	2%	WB	7.3	10.3	15.9	18.8	23.1	26.4	28.5	27.3	24.6	20.2	14.5	10.5	(29)	
		MCDB	10.7	15.2	20.0	25.2	27.9	31.9	33.7	32.0	30.6	24.2	18.6	12.1	(30)	
	5%	WB	5.2	7.5	14.2	16.9	21.5	25.1	27.3	26.2	23.5	18.6	12.1	7.2	(31)	
		MCDB	8.5	11.4	18.5	22.8	26.8	30.5	32.7	31.0	28.8	22.6	16.5	10.0	(32)	
	10%	WB	3.2	5.1	11.4	15.2	20.0	24.0	26.3	25.1	22.3	16.4	9.9	4.8	(33)	
		MCDB	5.7	8.1	16.6	20.7	24.9	29.1	31.5	29.5	26.8	21.5	13.9	7.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.7	12.0	13.0	11.6	11.3	11.3	11.4	13.6	13.5	12.0	10.3	(35)	
		MCDBR	14.9	15.6	17.7	17.5	14.8	12.7	12.6	13.1	15.0	16.9	16.6	14.6	(36)	
		MCWBR	10.9	10.5	10.4	9.3	6.7	5.9	6.7	7.2	7.1	8.9	10.4	10.6	(37)	
	5% WB	MCDBR	14.2	14.6	14.3	14.0	12.1	11.4	11.4	11.8	12.9	12.9	15.0	13.1	(38)	
		MCWBR	10.6	10.4	9.3	8.6	6.6	6.3	6.8	7.3	6.9	8.2	10.3	10.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.280	0.315	0.326	0.382	0.410	0.416	0.427	0.420	0.377	0.334	0.297	0.285	(40)	
	taud		2.468	2.380	2.415	2.278	2.268	2.300	2.260	2.297	2.370	2.478	2.511	2.464	(41)	
	Ebn at Noon		890	897	927	889	868	861	847	843	859	867	862	851	(42)	
	Edh at Noon		76	97	105	129	133	130	134	125	109	87	73	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.07	2.97	3.87	4.89	5.80	6.44	6.46	5.64	4.68	3.27	2.28	1.74	(44)	
	RadStd		0.16	0.20	0.35	0.31	0.48	0.51	0.45	0.32	0.31	0.38	0.18	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page